

4. Sentiment analysis

```
In [ ]: # ! pip install torch
```

```
In [ ]: import nltk
import tensorflow as tf
from tensorflow.keras.layers import Dense, LSTM
from tensorflow.keras.models import Sequential
import plotly.graph_objects as go
from plotly.subplots import make_subplots
```

```
In [ ]: import pandas as pd
import json
import pandas as pd
import numpy as np
import plotly as plt
import warnings
warnings.filterwarnings('ignore')
pd.options.plotting.backend = "plotly"
# import yfinance as yf
```

Step1&2: get the Processed data

```
In [ ]: topic=pd.read_excel("topic modeling in processed data.xlsx")
topic.head(2)
```

```
Out[ ]:
```

	publish time	headline_text	Sentence_length	doc	words	words_no_stop	words_final	words
0	2023-01-02	Hedge Fund Align Urges Korean Banks to Boost S...	64	hedge fund align urges korean banks to boost s...	['hedge', 'fund', 'align', 'urges', 'korean', ...	['hedge', 'fund', 'align', 'urges', 'korean', ...	['hedg', 'fund', 'align', 'urg', 'korean', 'ba...	[(0, 1, (3, 1),
1	2023-01-02	Israel's Smotrich May Adopt Rules Forcing Disc...	63	israels smotrich may adopt rules forcing disco...	['israels', 'smotrich', 'may', 'adopt', 'rules...	['israels', 'smotrich', 'may', 'adopt', 'rules...	['israel', 'smotrich', 'may', 'adopt', 'rule',...	[(9, 1), (12,

```
In [ ]: fin_svb=pd.read_csv(".\data source\SIVBQ_20230102_20230331.csv")
fin_svb['Date']=pd.to_datetime(fin_svb['Date'])
fin_svb.set_index('Date',inplace=True)
fin_svb.head()
```

Out []:

	Open	High	Low	Close	Adj Close	Volume
Date						
2023-01-03	232.169998	235.000000	221.649994	225.220001	225.220001	764645
2023-01-04	230.100006	241.632507	228.710007	240.059998	240.059998	854687
2023-01-05	235.710007	237.389999	222.410004	232.589996	232.589996	985262
2023-01-06	237.119995	248.190002	231.429993	245.789993	245.789993	1017532
2023-01-09	247.960007	254.940002	244.529999	249.429993	249.429993	1102929

In []:

```
data=pd.read_excel("news_titles_with_attitude.xlsx")
data.head(2)
```

Out []:

	publish time	headline_text	Sentence_length	doc	words	words_no_stop	words_final	words
0	2023-01-02	Hedge Fund Align Urges Korean Banks to Boost S...	64	hedge fund align urges korean banks to boost s...	['hedge', 'fund', 'align', 'urges', 'korean', 'banks', 'to', 'boost', 's...']	['hedge', 'fund', 'align', 'urges', 'korean', ...]	['hedg', 'fund', 'align', 'urg', 'korean', 'ba...']	[(0, 1), (3, 1), ...]
1	2023-01-02	Israel's Smotrich May Adopt Rules Forcing Disc...	63	israels smotrich may adopt rules forcing disco...	['israels', 'smotrich', 'may', 'adopt', 'rules', 'forcing', 'disco...']	['israels', 'smotrich', 'may', 'adopt', 'rules...']	['israel', 'smotrich', 'may', 'adopt', 'rule',...]	[(9, 1), (12, 1), ...]

In []:

```
dt = data[ ['publish time', 'Sentence_length', 'headline_text', 'words_final', 'attitude']]
dt['Topic'] = topic[ ['Topic']]
dt.head(2)
```

Out []:

	publish time	Sentence_length	headline_text	words_final	attitude	Topic
0	2023-01-02	64	Hedge Fund Align Urges Korean Banks to Boost S...	['hedg', 'fund', 'align', 'urg', 'korean', 'ba...']	0.0	3
1	2023-01-02	63	Israel's Smotrich May Adopt Rules Forcing Disc...	['israel', 'smotrich', 'may', 'adopt', 'rule',...]	0.0	3

In []:

```
dt [ dt['attitude']>0 ].count()
```

Out []:

publish time	1335	Sentence_length	1335	headline_text	1335	words_final	1335	attitude	1335	Topic	1335
final		1335	attitude		1335	Topic		1335	dtype: int64		

In []:

```
dt [ dt['attitude']==0 ].count()
```

Out []:

publish time	6021	Sentence_length	6021	headline_text	6021	words_final	6021	attitude	6021	Topic	6021
final		6021	attitude		6021	Topic		6021	dtype: int64		

In []:

```
svb_dates = fin_svb.index
all_dates = dt['publish time'].unique()

m=[]
for d in all_dates:
```

```

    if d in svb_dates:
        m.append( round(fin_svb.Close.loc[d], 2) )
    else:
        m.append(np.nan)

svb = pd.DataFrame(index=all_dates)
# svb['date'] = all_dates

svb['Close'] = m
svb.fillna(axis=0, method="ffill", inplace=True)
print(svb)
svb.plot(
    kind='line',
    title='SVB stock price',
)

```

```

                Close2023-01-02      NaN2023-01-03  225.222023-01-04  240.06202
3-01-05  232.592023-01-06  245.79...          ...2023-03-27  106.042023-0
3-28    0.402023-03-29    0.972023-03-30    0.892023-03-31    0.89[88 rows
x 1 columns]

```

```

In [ ]: svb_dates = svb.index
all_dates = dt['publish time'].tolist()

m=[]
for d in all_dates:
    if d in svb_dates:
        m.append( round(svb.Close.loc[d], 2) )
    else:
        m.append(np.nan)

dt['svb_close_price'] = m
dt.loc[20:40]

```

Out[]:

	publish time	Sentence_length	headline_text	words_final	attitude	Topic	svb_close_price
20	2023-01-03	52	Citizens Financial Cut to Neutral at Wedbush; ...	['citizen', 'financi', 'cut', 'neutral', 'wedb...	0.000000	6	225.22
21	2023-01-03	62	Bank of China A Shares Raised to Add at CGS- CI...	['bank', 'china', 'share', 'rais', 'add', 'cgs...	0.000000	3	225.22
22	2023-01-03	52	MANDATE: BNG Bank EUR Benchmark 10Y Social Bon...	['mandat', 'bng', 'bank', 'eur', 'benchmark', ...	0.033333	9	225.22
23	2023-01-03	59	Citic Bank A Shares Raised to Add at CGS- CIMB;...	['citic', 'bank', 'share', 'rais', 'add', 'cgs...	0.000000	6	225.22
24	2023-01-03	52	Bocom H Shares Raised to Add at CGS- CIMB; PT H...	['bocom', 'h', 'share', 'rais', 'add', 'cgscim...	0.000000	6	225.22
25	2023-01-03	54	Bocom A Shares Raised to Add at CGS- CIMB; PT 5...	['bocom', 'share', 'rais', 'add', 'cgscimb', '...]	0.000000	6	225.22
26	2023-01-03	49	M&T Bank Raised to Outperform at Wedbush; PT \$170	['mt', 'bank', 'rais', 'outperform', 'wedbush'...	0.000000	6	225.22
27	2023-01-03	59	Romania Dec. International Reserves Rise to EU...	['romania', 'dec', 'intern', 'reserv', 'rise',...	0.000000	2	225.22
28	2023-01-03	63	Credit Suisse's Cathal Deasy to Leave Bank, Fi...	['credit', 'suiss', 'cathal', 'deasi', 'leav', '...	0.000000	2	225.22
29	2023-01-03	50	*MANDATE: WESTPAC £BENCHMARK 5Y SONIA COVERED ...	['mandat', 'westpac', 'benchmark', 'sonia', 'c...	0.000000	9	225.22
30	2023-01-03	64	Brazil Media: Gasoline Exemption; Port Termina...	['brazil', 'media', 'gasolin', 'exempt', 'port...	0.000000	8	225.22
31	2023-01-03	55	Taiwan to Widen Foreign Banks' Use of TWD Bond...	['taiwan', 'widen', 'foreign', 'bank', 'use', ...]	-0.125000	5	225.22
32	2023-01-03	52	MANDATE: Westpac GBP Benchmark 5Y	['mandat', 'westpac', 'gbp', 'benchmark',	0.000000	9	225.22

	publish time	Sentence_length	headline_text	words_final	attitude	Topic	svb_close_price
			Sonia Covere...	'son...			
33	2023-01-03	62	Citizens Financial Cut to Neutral at Webush on...	['citizen', 'financi', 'cut', 'neutral', 'webu...	0.000000	6	225.22
34	2023-01-03	55	MANDATE: Aareal Bank EUR Benchmark 10/2027 Cov...	['mandat', 'aareal', 'bank', 'eur', 'benchmark...	0.000000	9	225.22
35	2023-01-03	56	*MANDATE: AAREAL BANK EUR BENCHMARK 10/2027 CO...	['mandat', 'aareal', 'bank', 'eur', 'benchmark...	0.000000	9	225.22
36	2023-01-03	54	PRICED: Walliser Kantonalbank CHF220m 5.5Y SAR...	['price', 'wallis', 'kantonalbank', 'chfm', 's...	0.000000	9	225.22
37	2023-01-03	58	GUIDANCE: Swedish Export Credit GBP Bmark Long...	['guidanc', 'swedish', 'export', 'credit', 'gb...	-0.050000	9	225.22
38	2023-01-03	47	GUIDANCE: KfW GBP Benchmark Long 3Y UKT +81 Area	['guidanc', 'kfw', 'gbp', 'benchmark', 'long',...	-0.050000	9	225.22
39	2023-01-03	44	*GUIDANCE: SEK GBP BMARK 3/2026 UKT+105 AREA	['guidanc', 'sek', 'gbp', 'bmark', 'ukt', 'area']	0.000000	9	225.22
40	2023-01-03	60	*BOE ALLOTS GBP 150MLN IN LIQUIDITY OP, MATCHI...	['boe', 'allot', 'gbp', 'mln', 'liquid', 'op',...	0.000000	4	225.22

Step 3: Feature extraction

Word2Vec

```
In [ ]: dt["sentiment"] = [[1,0] if x>0 else [0,1] for x in dt["attitude"]]
```

```
In [ ]: # choose headlines when headline exist
dt=dt[dt["Sentence_length"]>0]
```

```
In [ ]: from gensim.models import Word2Vec
model = Word2Vec(dt['words_final'], vector_size=50, window=2, min_count=1, sg=0)
w2v_weights=model.wv.vectors
vocab_size, embedding_size = w2v_weights.shape
print("Vocabulary Size: {} - Embedding Dim: {}".format(vocab_size, embedding_size))
```

Vocabulary Size: 37 - Embedding Dim: 50

```
In [ ]: xreview=[]
for l in dt.index:
```

```

doc=[w for w in dt.loc[l,"words_final"] if w in model.wv.index_to_key]
if len(doc)>0:
    xreview.append(model.wv[doc])
else:
    print(l,doc)
    xreview.append(np.zeros(model.vector_size))
dt["w2v"]=xreview

```

check

```
In [ ]: dt.head(2)
```

```
Out [ ]:
```

	publish time	Sentence_length	headline_text	words_final	attitude	Topic	svb_close_price	sentiment
0	2023-01-02	64	Hedge Fund Align Urges Korean Banks to Boost S...	['hedg', 'fund', 'align', 'urg', 'korean', 'ba...	0.0	3	NaN	[0, 1]
1	2023-01-02	63	Israel's Smotrich May Adopt Rules Forcing Disc...	['israel', 'smotrich', 'may', 'adopt', 'rule',...	0.0	3	NaN	[0, 1]

```
In [ ]: max_length = 0
min_length=10000
for row in dt['w2v']:
    if len(row) < min_length:
        min_length = len(row)
    if len(row) > max_length:
        max_length = len(row)
print(min_length,max_length)
```

23 95

check

```
In [ ]: #for short sentences, we prepend zero padding so all input to RNN has same length

padded_w2v = []
for v in dt["w2v"]:
    zero_padding_cnt = max_length - v.shape[0]
    pad = np.zeros((1, 50))
    for i in range(zero_padding_cnt):
        v = np.concatenate((pad, v), axis=0)
    padded_w2v.append(v)

dt["padded_w2v"]=padded_w2v

```

```
In [ ]: dt.head(2)
```

Out[]:	publish time	Sentence_length	headline_text	words_final	attitude	Topic	svb_close_price	sentiment
0	2023-01-02	64	Hedge Fund Align Urges Korean Banks to Boost S...	['hedg', 'fund', 'align', 'urg', 'korean', 'ba...]	0.0	3	NaN	[0, 1]
1	2023-01-02	63	Israel's Smotrich May Adopt Rules Forcing Disc...	['israel', 'smotrich', 'may', 'adopt', 'rule',...]	0.0	3	NaN	[0, 1]

Step 4: Spilt dataset into train and test dataset

```
In [ ]: from torch import randperm

np.random.seed(42) # Set the random seed to 42

len(dt)
meaningless_number = randperm(len(dt)).tolist() # 生成乱序的索引
dt['meaningless_number'] = meaningless_number
dt.sort_values(by='meaningless_number', inplace=True)
dt
```

Out[]:

	publish time	Sentence_length	headline_text	words_final	attitude	Topic	svb_close_price	sentin
2572	2023-02-02	62	Santander Polska Expects Net Interest Margin t...	['santand', 'polska', 'expect', 'net', 'intere...]	0.0	1	333.50	
7772	2023-03-29	55	Dutch Trade Union Plans Another Strike at ING ...	['dutch', 'trade', 'union', 'plan', 'anoth', '...]	0.0	0	0.97	
2286	2023-01-30	62	MANDATE: Arbejdernes Landsbank SEK and/or NOK ...	['mandat', 'arbejdern', 'landsbank', 'sek', 'a...]	0.0	9	293.96	
5566	2023-03-10	63	ASIC: ANZ Penalized A \$10M Over Home Loan Intro...	['asic', 'anz', 'penal', 'home', 'loan', 'intr...]	0.0	3	106.04	
5809	2023-03-13	56	Banks' Plunges and Trading Halts Show Crisis N...	['bank', 'plung', 'trade', 'halt', 'show', 'cr...]	0.0	7	106.04	
...	...	...	...	...	...	...	...	...
4978	2023-03-03	64	Hungary Central Bank Dismisses Bank Plea to Sc...	['hungari', 'central', 'bank', 'dismiss', 'ban...]	0.0	3	284.41	
3180	2023-02-08	64	Mauritius's SBM Names Nilamber as Group Chief ...	['mauritius', 'sbm', 'name', 'nilamb', 'group...]	0.0	4	320.40	
4188	2023-02-21	50	Mizuho Trust Fires Employee For Embezzling 60m...	['mizuho', 'trust', 'fire', 'employe', 'embezz...]	0.0	9	285.71	
3126	2023-02-08	64	Goldman Sachs Names Banker in Senior Americas ...	['goldman', 'sach', 'name', 'banker', 'senior'...]	0.0	6	320.40	
8098	2023-03-31	37	NAB Cut to Reduce at CLSA; PT A \$27.90	['nab', 'cut', 'reduc', 'clsa', 'pt']	0.0	6	0.89	

8137 rows × 11 columns


```
In [ ]: # spilt the df:dt into train an dtest dataset
train=dt.iloc[:int(0.7*len(dt))]
test=dt.iloc[int(0.7*len(dt)):]

train.sort_values(by='publish time', ascending=True, inplace=True)
train.drop(columns='meaningless_number', inplace=True)
test.sort_values(by='publish time', ascending=True, inplace=True)
test.drop(columns='meaningless_number', inplace=True)
```

```
In [ ]: dt.sort_values(by='publish time', ascending=True, inplace=True)
dt.drop(columns='meaningless_number', inplace=True)
```

check

```
In [ ]: dt
```

Out[]:

	publish time	Sentence_length	headline_text	words_final	attitude	Topic	svb_close_price
8	2023-01-02	45	MANDATE: BPCE EUR Benchmark SNP; Long 5Y, 10Y	['mandat', 'bpce', 'eur', 'benchmark', 'snp', ...	-0.050000	9	NaN
4	2023-01-02	61	Actis Is Said to Be in Talks to Refinance Min....	['acti', 'said', 'talk', 'refin', 'min', 'inrb...	0.000000	0	NaN
14	2023-01-02	60	*GUIDANCE: ERSTE GROUP BANK €BENCHMARK 6Y COVE...	['guidanc', 'erst', 'group', 'bank', 'benchmar...	0.000000	9	NaN
13	2023-01-02	51	GUIDANCE: Erste EUR Benchmark 6Y Covered MS +23...	['guidanc', 'erst', 'eur', 'benchmark', 'cover...	0.000000	9	NaN
6	2023-01-02	64	*NEW DEAL:EDELWEISS FINANCIAL PLANS UP TO INR4...	['new', 'dealedelweiss', 'financi', 'plan', 'i...	0.045455	0	NaN
...	...	...	...	...	...	...	...
8012	2023-03-31	63	UAE Cenbank Revokes the Licence of MTS Bank Br...	['uae', 'cenbank', 'revok', 'licenc', 'mt', 'b...	0.000000	7	0.89
8030	2023-03-31	61	Fed Rate Cut Would Lead to Stagflation, Instab...	['fed', 'rate', 'cut', 'lead', 'stagflat', 'in...	0.000000	7	0.89
8079	2023-03-31	38	Poland's mBank Appoints Ruhland as CFO	['poland', 'mbank', 'appoint', 'ruhland', 'cfo']	0.000000	7	0.89
8041	2023-03-31	63	US LEVLOANS CALENDAR: Melissa & Doug's A&E Ter...	['us', 'levloan', 'calendar', 'melissa', 'doug...	-0.125000	7	0.89
8098	2023-03-31	37	NAB Cut to Reduce at CLSA; PT A\$27.90	['nab', 'cut', 'reduc', 'clsa', 'pt']	0.000000	6	0.89

8137 rows × 10 columns

In []:

test

Out[]:

	publish time	Sentence_length	headline_text	words_final	attitude	Topic	svb_close_price
5	2023-01-02	64	NEW DEAL: Edelweiss Financial Plans Up to INR4...	['new', 'deal', 'edelweiss', 'financi', 'plan'...	0.045455	0	NaN
15	2023-01-02	56	Paschi Says It Has Overcome Risks to Business ...	['paschi', 'say', 'overcom', 'risk', 'busi', '...'...	0.000000	7	NaN
6	2023-01-02	64	*NEW DEAL:EDELWEISS FINANCIAL PLANS UP TO INR4...	['new', 'dealedelweiss', 'financi', 'plan', 'i...'...	0.045455	0	NaN
11	2023-01-02	59	*MANDATE: BERLIN HYP €1B COVERED; LONG 3Y SOCI...	['mandat', 'berlin', 'hyp', 'b', 'cover', 'lon...'...	-0.072222	9	NaN
4	2023-01-02	61	Actis Is Said to Be in Talks to Refinance Min....	['acti', 'said', 'talk', 'refin', 'min', 'inrb...'...	0.000000	0	NaN
...	...	...	...	...	...	...	...
8125	2023-03-31	60	*AUSTRALIA FEB. CREDIT TO BUSINESS, CONSUMERS ...	['australia', 'feb', 'credit', 'busi', 'consum...'...	0.000000	2	0.89
8052	2023-03-31	45	*PHILIPPINES FEB. BANK NET LOANS RISE 10% Y/Y	['philippin', 'feb', 'bank', 'net', 'loan', 'r...'...	0.000000	2	0.89
8130	2023-03-31	59	JBIC, Others Give Vietcombank \$300m Loan for G...	['jbic', 'other', 'give', 'vietcombank', 'm', '...'...	-0.200000	9	0.89
8134	2023-03-31	53	*SMFG SOUNDS OUT INVESTORS ABOUT AT1 YEN BOND ...	['smfg', 'sound', 'investor', 'yen', 'bond', '...'...	0.000000	5	0.89
8098	2023-03-31	37	NAB Cut to Reduce at CLSA; PT A\$27.90	['nab', 'cut', 'reduc', 'clsa', 'pt']	0.000000	6	0.89

2442 rows × 10 columns

In []: train

Out[]:

	publish time	Sentence_length	headline_text	words_final	attitude	Topic	svb_close_price	sent
12	2023-01-02	61	Paschi Jumps; Bank Says Risks to Business Cont...	['paschi', 'jump', 'bank', 'say', 'risk', 'bus...	0.000000	7	NaN	
10	2023-01-02	63	MANDATE: Berlin Hyp EU1b WNG Covered; Long 3Y ...	['mandat', 'berlin', 'hyp', 'eub', 'wng', 'cov...	-0.072222	9	NaN	
14	2023-01-02	60	*GUIDANCE: ERSTE GROUP BANK €BENCHMARK 6Y COVE...	['guidanc', 'erst', 'group', 'bank', 'benchmar...	0.000000	9	NaN	
0	2023-01-02	64	Hedge Fund Align Urges Korean Banks to Boost S...	['hedg', 'fund', 'align', 'urg', 'korean', 'ba...	0.000000	3	NaN	
8	2023-01-02	45	MANDATE: BPCE EUR Benchmark SNP; Long 5Y, 10Y	['mandat', 'bpce', 'eur', 'benchmark', 'snp', ...	-0.050000	9	NaN	
...	...	...	...	...	...	...	...	...
8039	2023-03-31	56	Dometic Renews Credit Facility Agreement With ...	['domet', 'renew', 'credit', 'facil', 'agreeme...	0.000000	3	0.89	
8005	2023-03-31	58	Storebrand Raised to Buy at Kepler Cheuvreux; ...	['storebrand', 'rais', 'buy', 'kepler', 'cheuv...	0.000000	6	0.89	
8058	2023-03-31	60	China Construction Bank Loan Growth to Support...	['china', 'construct', 'bank', 'loan', 'growth...	0.000000	3	0.89	
8089	2023-03-31	62	*COLLINS: STILL SEEING QUITE A BIT OF STRENGTH...	['collin', 'still', 'see', 'quit', 'bit', 'str...	0.000000	4	0.89	
8011	2023-03-31	49	Vietnamese Banks' Lending Rose 2.1% From End-2022	['vietnames', 'bank', 'lend', 'rose', 'end']	0.600000	3	0.89	

5695 rows × 10 columns

```
In [ ]: test["sentiment"].value_counts()
```

```
Out[ ]: sentiment[0, 1]      2033[1, 0]      409Name: count, dtype: int64
```

```
In [ ]: train["sentiment"].value_counts()
```

```
Out[ ]: sentiment[0, 1]      4769[1, 0]      926Name: count, dtype: int64
```

```
In [ ]: X_train=np.array(train["padded_w2v"].tolist())
X_test=np.array(test["padded_w2v"].tolist())
Y_train=np.array(train["sentiment"].tolist())
Y_test=np.array(test["sentiment"].tolist())
```

Step 5: Modeling

```
In [ ]: # LSTM model
model = Sequential()
model.add(LSTM(32))
model.add(Dense(2, activation='softmax'))
model.compile(loss='categorical_crossentropy',
              optimizer='adam',
              metrics=['accuracy'])
```

```
In [ ]: print('Train...')
model.fit(X_train, Y_train, epochs=10)
```

```
Train...Epoch 1/10178/178 [=====] - 8s 29ms/step -
  loss: 0.4605 - accuracy: 0.8195Epoch 2/10178/178 [=====
=====] - 5s 29ms/step - loss: 0.4434 - accuracy: 0.8374Epoch 3/10178/178 [
=====] - 6s 31ms/step - loss: 0.4421 - accuracy: 0
.8374Epoch 4/10178/178 [=====] - 5s 28ms/step - lo
ss: 0.4396 - accuracy: 0.8374Epoch 5/10178/178 [=====
===] - 5s 27ms/step - loss: 0.4347 - accuracy: 0.8374Epoch 6/10178/178 [===
=====] - 5s 28ms/step - loss: 0.4319 - accuracy: 0.83
78Epoch 7/10178/178 [=====] - 5s 27ms/step - loss:
  0.4286 - accuracy: 0.8390Epoch 8/10178/178 [=====
] - 5s 27ms/step - loss: 0.4283 - accuracy: 0.8409Epoch 9/10178/178 [=====
=====] - 5s 30ms/step - loss: 0.4231 - accuracy: 0.8435E
poch 10/10178/178 [=====] - 5s 29ms/step - loss: 0
.4205 - accuracy: 0.8434
```

```
Out[ ]: <keras.callbacks.History at 0x1e506745f10>
```

```
In [ ]: score, acc = model.evaluate(X_train, Y_train, verbose=2)
print('Train score:', score)# Loss value
print('Train accuracy:', acc)
```

```
178/178 - 2s - loss: 0.4181 - accuracy: 0.8451 - 2s/epoch - 13ms/stepTrain
score: 0.41813361644744873Train accuracy: 0.845127284526825Train score: 0.4
1813361644744873Train accuracy: 0.845127284526825
```

```
In [ ]: score, acc = model.evaluate(X_test, Y_test, verbose=2)
print('Test score:', score)
print('Test accuracy:', acc)
```

```
77/77 - 1s - loss: 0.4404 - accuracy: 0.8370 - 867ms/epoch - 11ms/stepTest
score: 0.44037100672721863Test accuracy: 0.8370188474655151Test score: 0.44
037100672721863Test accuracy: 0.8370188474655151
```

```
In [ ]: temp=model.predict(X_test)
test["pred"]=[1 if x[0]>x[1] else 0 for x in list(temp)]
```

```
77/77 [=====] - 1s 11ms/step
```

```
In [ ]: temp=model.predict(X_test)
test["pred"]=[1 if x[0]>x[1] else 0 for x in list(temp)]
```

```

confusion_matrix=pd.DataFrame()
confusion_matrix["realY"]=[1 if x[0]==1 else 0 for x in test["sentiment"].values]
confusion_matrix["predictY"]=test["pred"].values
pd.crosstab(confusion_matrix["predictY"],confusion_matrix["realY"])

```

77/77 [=====] - 1s 13ms/step

Out []: **realY** **0** **1**

predictY

0 2032 397

1 1 12

```

In [ ]: test["prob1"]=[x[0] for x in list(temp)]
test["prob_fast"]=(test["prob1"]).rolling(100).mean()
test["prob_slow"]=(test["prob1"]).rolling(50).mean()

```

check

Step 6: Model result

```
In [ ]: train.head(2)
```

Out []: **publish**
 time **Sentence_length** **headline_text** **words_final** **attitude** **Topic** **svb_close_price** **sentime**

12	2023-01-02	61	Paschi Jumps; Bank Says Risks to Business Cont...	['paschi', 'jump', 'bank', 'say', 'risk', 'bus...	0.000000	7	NaN	[C
-----------	------------	----	---	---	----------	---	-----	----

10	2023-01-02	63	MANDATE: Berlin Hyp EU1b WNG Covered; Long 3Y ...	['mandat', 'berlin', 'hyp', 'eub', 'wng', 'cov...	-0.072222	9	NaN	[C
-----------	------------	----	---	---	-----------	---	-----	----

```
In [ ]: test.head(2)
```

Out []: **publish**
 time **Sentence_length** **headline_text** **words_final** **attitude** **Topic** **svb_close_price** **sentime**

5	2023-01-02	64	NEW DEAL: Edelweiss Financial Plans Up to INR4...	['new', 'deal', 'edelweiss', 'financi', 'plan'...	0.045455	0	NaN	[1,
----------	------------	----	---	---	----------	---	-----	-----

15	2023-01-02	56	Paschi Says It Has Overcome Risks to Business ...	['paschi', 'say', 'overcom', 'risk', 'busi', '...	0.000000	7	NaN	[0,
-----------	------------	----	---	---	----------	---	-----	-----

Results statistics in test dataset

```

In [ ]: test_gp = test.groupby(['publish time']).agg({
    'headline_text': [len],
    'pred': [np.mean, np.max],
    'prob1': [np.mean],
    'prob_fast': [np.mean],

```

```

    'prob_slow': [np.mean],
    })
test_gp.head(20)

```

Out[]:

	headline_text		pred		prob1	prob_fast	prob_slow
	len	mean	amax	mean	mean	mean	
publish time							
2023-01-02	10	0.000000	0	0.168908	NaN	NaN	
2023-01-03	24	0.000000	0	0.110893	NaN	NaN	
2023-01-04	37	0.000000	0	0.104767	NaN	0.107124	
2023-01-05	27	0.000000	0	0.145302	NaN	0.116920	
2023-01-06	14	0.000000	0	0.153300	0.123575	0.137201	
2023-01-07	1	0.000000	0	0.235837	0.124698	0.146507	
2023-01-09	35	0.000000	0	0.117893	0.125639	0.137293	
2023-01-10	33	0.000000	0	0.107437	0.127695	0.114818	
2023-01-11	36	0.027778	1	0.142023	0.123428	0.122617	
2023-01-12	23	0.000000	0	0.124029	0.125480	0.138374	
2023-01-13	42	0.000000	0	0.121159	0.128482	0.127042	
2023-01-14	10	0.000000	0	0.122359	0.128164	0.119608	
2023-01-16	28	0.000000	0	0.141387	0.126174	0.123884	
2023-01-17	31	0.000000	0	0.125272	0.126595	0.131901	
2023-01-18	35	0.000000	0	0.131071	0.129990	0.132549	
2023-01-19	42	0.023810	1	0.119434	0.132194	0.131228	
2023-01-20	41	0.000000	0	0.123373	0.125551	0.117652	
2023-01-21	8	0.000000	0	0.099466	0.121211	0.120680	
2023-01-22	1	0.000000	0	0.130754	0.119229	0.119696	
2023-01-23	34	0.000000	0	0.138618	0.120086	0.123054	

In []:

```

test_gp_hd_len = test_gp['headline_text']
test_gp_hd_len

```

Out[]: **len**

publish time	
2023-01-02	10
2023-01-03	24
2023-01-04	37
2023-01-05	27
2023-01-06	14
...	...
2023-03-27	40
2023-03-28	50
2023-03-29	43
2023-03-30	34
2023-03-31	42

84 rows × 1 columns

```
In [ ]: test_gp_prob_fast = test_gp['prob_fast']
        test_gp_prob_fast
```

Out[]: **mean**

publish time	
2023-01-02	NaN
2023-01-03	NaN
2023-01-04	NaN
2023-01-05	NaN
2023-01-06	0.123575
...	...
2023-03-27	0.145289
2023-03-28	0.146472
2023-03-29	0.139793
2023-03-30	0.134840
2023-03-31	0.138581

84 rows × 1 columns

Result visualization in test dataset

```
In [ ]: svb.plot(
        kind='line',
        title='SVB stock price',
    )
```



```
In [ ]: # Create figure with secondary y-axis
fig = make_subplots(specs=[[{"secondary_y": True}]]))
r=4
# Add traces
fig.add_trace(
    go.Scatter(x=test['publish time'], y=test.svb_close_price, name='SVB stock price',

fig.add_trace(
    go.Scatter(x=test_gp_prob_fast.index, y=test_gp_prob_fast['mean'], name='Prob_fast(

# Add figure title
fig.update_layout(
    title_text="Trending of Positive Sentiment ",
    width=1200,
    height=600,
)

# Set x-axis title
fig.update_xaxes(title_text="Date")

# Set y-axes titles
fig.update_yaxes(title_text="SVB stock price ", secondary_y=False)
fig.update_yaxes(title_text="<b>Prob of Positive Sentiment</b>", secondary_y=True)

fig.show()
```

```
In [ ]: # Create figure with secondary y-axis
fig = make_subplots(specs=[[{"secondary_y": True}]]))
r=4
# Add traces
fig.add_trace(
    go.Scatter(x=test['publish time'], y=test['svb_close_price'], name='SVB stock price

fig.add_trace(
    go.Bar(x=test_gp_hd_len.index, y=test_gp_hd_len['len'], name='the number of daily h

# Add figure title
fig.update_layout(
    title_text="Trending of Positive Sentiment ",
    width=1200,
    height=600,
)

# Set x-axis title
fig.update_xaxes(title_text="Date")

# Set y-axes titles
fig.update_yaxes(title_text="SVB stock price ", secondary_y=False)
fig.update_yaxes(title_text="<b>Prob of Positive Sentiment</b>", secondary_y=True)

fig.show()
```

5. Sentiment result and Topic modeling result

topic modeling result

```
In [ ]: def count_number_of_headline_within_top_details(top_n_index=0,top_details_name=0,top_de
    a = top_details_name
    b = (top_n_index+1)
    c = top_details
    print('the headline number with topic',a,'ranked#',b,':',c)

top5=dt['Topic'].value_counts().index[0:5]

top_details = dt['Topic'].value_counts().tolist()
```

```
len_top_details = len(top_details)
top_details_name = dt['Topic'].value_counts().index.tolist()
```

```
In [ ]: print(len_top_details,'topics are generated here')
print('-----')
print(top_details)
print(top_details_name)
print('-----')
for i in range(len_top_details):
    count_number_of_headline_within_top_details(i,top_details_name[i],top_details[i])

10 topics are generated here-----[1210
, 1169, 1089, 1035, 750, 742, 604, 567, 505, 466][7, 1, 9, 6, 3, 0, 5, 2, 4
, 8]-----the headline number with topic
7 ranked# 1 : 1210the headline number with topic 1 ranked# 2 : 1169the head
ine number with topic 9 ranked# 3 : 1089the headline number with topic 6 ran
ked# 4 : 1035the headline number with topic 3 ranked# 5 : 750the headline num
ber with topic 0 ranked# 6 : 742the headline number with topic 5 ranked# 7 :
604the headline number with topic 2 ranked# 8 : 567the headline number with
topic 4 ranked# 9 : 505the headline number with topic 8 ranked# 10 : 466
```

topic=topic7

Results statistics in test dataset

```
In [ ]: test1 = test[test['Topic']==top5[0]]
```

```
In [ ]: test_gp1 = test1.groupby(['publish time']).agg({
    'headline_text':[len],
    'pred':[np.mean,np.max],
    'prob1': [np.mean],
    'prob_fast': [np.mean],
    'prob_slow': [np.mean],
})
test_gp1.head()
```

```
Out[ ]:
```

	headline_text		pred	prob1	prob_fast	prob_slow
	len	mean	amax	mean	mean	mean
publish time						
2023-01-02	1	0.0	0	0.213150	NaN	NaN
2023-01-03	1	0.0	0	0.131326	NaN	NaN
2023-01-04	3	0.0	0	0.199756	NaN	0.106729
2023-01-05	4	0.0	0	0.167288	NaN	0.118911
2023-01-06	2	0.0	0	0.160661	0.123647	0.138895

```
In [ ]: test_gp1_hd_len = test_gp1['headline_text']
test_gp1_hd_len
```

Out[]: **len**

publish time	
2023-01-02	1
2023-01-03	1
2023-01-04	3
2023-01-05	4
2023-01-06	2
...	...
2023-03-27	7
2023-03-28	10
2023-03-29	8
2023-03-30	9
2023-03-31	5

71 rows × 1 columns

```
In [ ]: test_gpl_prob_fast = test_gpl['prob_fast']
        test_gpl_prob_fast
```

Out[]: **mean**

publish time	
2023-01-02	NaN
2023-01-03	NaN
2023-01-04	NaN
2023-01-05	NaN
2023-01-06	0.123647
...	...
2023-03-27	0.145485
2023-03-28	0.146278
2023-03-29	0.140006
2023-03-30	0.134599
2023-03-31	0.138916

71 rows × 1 columns

Result visualization in test dataset

```
In [ ]: i=0
        count_number_of_headline_within_top_details(i,top_details_name[i],top_details[i])

        the headline number with topic 7 ranked# 1 : 1210
```

```
In [ ]: # Create figure with secondary y-axis
        fig = make_subplots(specs=[[{"secondary_y": True}]])
```

```

r=4
# Add traces
fig.add_trace(
    go.Scatter(x=test['publish time'], y=test.svb_close_price, name='SVB stock price',

fig.add_trace(
    go.Scatter(x=test_gpl_prob_fast.index, y=test_gpl_prob_fast['mean'], name='Prob_fas

# Add figure title
fig.update_layout(
    title_text="Trending of Positive Sentiment ",
    width=1200,
    height=600,
)

# Set x-axis title
fig.update_xaxes(title_text="Date")

# Set y-axes titles
fig.update_yaxes(title_text="SVB stock price ", secondary_y=False)
fig.update_yaxes(title_text="<b>Prob of Positive Sentiment</b>", secondary_y=True)

fig.show()

```

```

In [ ]: # Create figure with secondary y-axis
fig = make_subplots(specs=[[{"secondary_y": True}]]
r=4
# Add traces
fig.add_trace(
    go.Scatter(x=test['publish time'], y=test.svb_close_price, name='SVB stock price',

fig.add_trace(
    go.Bar(x=test_gpl_hd_len.index, y=test_gpl_hd_len['len'], name='the number of daily

# Add figure title
fig.update_layout(
    title_text="Trending of Positive Sentiment ",
    width=1200,
    height=600,
)

# Set x-axis title
fig.update_xaxes(title_text="Date")

# Set y-axes titles
fig.update_yaxes(title_text="SVB stock price ", secondary_y=False)
fig.update_yaxes(title_text="<b>Prob of Positive Sentiment</b>", secondary_y=True)

```

Experiment: Result visualization in whole dataset

```

In [ ]: wdt=dt
# wdt

w_X=np.array(wdt["padded_w2v"].tolist())

w_temp=model.predict(w_X)
wdt["pred"]=[1 if x[0]>x[1] else 0 for x in list(w_temp)]

wdt["prob1"]=[x[0] for x in list(w_temp)]
wdt["prob_fast"]=(wdt["prob1"]).rolling(100).mean()
wdt["prob_slow"]=(wdt["prob1"]).rolling(50).mean()

wdt_gp = wdt.groupby(['publish time']).agg({
    'headline_text': [len],
    'pred': [np.mean, np.max, np.min],

```

```

    'prob1': [np.mean],
    'prob_fast': [np.mean],
    'prob_slow': [np.mean],
    })
wdt_gp.head(10)

```

255/255 [=====] - 3s 11ms/step

Out []:

	headline_text			pred	prob1	prob_fast	prob_slow
	len	mean	amax	amin	mean	mean	mean
publish time							
2023-01-02	19	0.000000	0	0	0.156137	NaN	NaN
2023-01-03	99	0.000000	0	0	0.118204	0.122947	0.121389
2023-01-04	124	0.000000	0	0	0.111773	0.114810	0.113830
2023-01-05	78	0.000000	0	0	0.136503	0.120738	0.127780
2023-01-06	62	0.000000	0	0	0.135769	0.133506	0.133842
2023-01-07	7	0.000000	0	0	0.150160	0.134444	0.135891
2023-01-08	1	0.000000	0	0	0.151530	0.137729	0.137674
2023-01-09	104	0.000000	0	0	0.112700	0.124869	0.119238
2023-01-10	121	0.008264	1	0	0.116581	0.114041	0.115562
2023-01-11	101	0.009901	1	0	0.136738	0.125306	0.131039

```

In [ ]: wdt_gp_hd_len = wdt_gp['headline_text']
        # wdt_gp_hd_len

```

```

In [ ]: wdt_gp_prob_fast = wdt_gp['prob_fast']
        # wdt_gp_prob_fast

```

```

In [ ]: wdt_gp_prob_fast = wdt_gp['prob_fast']
        # wdt_gp_prob_fast

```

```

In [ ]: import plotly.graph_objects as go
        from plotly.subplots import make_subplots
        # Create figure with secondary y-axis
        fig = make_subplots(specs=[[{"secondary_y": True}]]
        r=4
        # Add traces
        fig.add_trace(
            go.Scatter(x=wdt['publish time'], y=wdt.svb_close_price, name='SVB stock price', mo
        fig.add_trace(
            go.Scatter(x=wdt['publish time'], y=wdt.prob_fast, name='Prob_fast(50) of Positive
        # Add figure title
        fig.update_layout(
            title_text="Trending of Positive Sentiment ",
            width=1200,
            height=600,
        )
        # Set x-axis title
        fig.update_xaxes(title_text="Date")
        # Set y-axes titles
        fig.update_yaxes(title_text="SVB stock price ", secondary_y=False)
        fig.update_yaxes(title_text="<b>Prob of Positive Sentiment</b>", secondary_y=True)
        fig.show()

```

```
In [ ]: # Create figure with secondary y-axis
fig = make_subplots(specs=[[{"secondary_y": True}]]
r=4
# Add traces
fig.add_trace(
    go.Scatter(x=wdt['publish time'], y=wdt['svb_close_price'], name='SVB stock price',

fig.add_trace(
    go.Bar(x=wdt_gp_hd_len.index, y=wdt_gp_hd_len['len'], name='the number of headline'

# Add figure title
fig.update_layout(
    title_text="Trending of Positive Sentiment ",
    width=1200,
    height=600,
)

# Set x-axis title
fig.update_xaxes(title_text="Date")

# Set y-axes titles
fig.update_yaxes(title_text="SVB stock price ", secondary_y=False)
fig.update_yaxes(title_text="<b>Prob of Positive Sentiment</b>", secondary_y=True)

fig.show()
```

*Export processed data

```
In [ ]: wdt.to_excel('sentiment analysis result(whole processed data) .xlsx')
```